5.6. Part 2 Demonstrate Knowledge Base and Query System in Prolog

https://swish.swi-prolog.org/

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% Medical Expert System
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% Patients

patient(rahul).
patient(priya).
patient(amit).
patient(neha).

% Diseases

disease(rahul, fever).
disease(priya, cough).
disease(amit, diabetes).
disease(neha, migraine).

% Symptoms

symptom(fever, high_temperature).
symptom(fever, headache).
symptom(cough, sore_throat).
symptom(cough, cold).
symptom(diabetes, frequent_urination).
symptom(diabetes, increased_thirst).
symptom(migraine, severe_headache).
symptom(migraine, nausea).

% Medicines

medicine(fever, paracetamol).
medicine(cough, cough_syrup).
medicine(diabetes, insulin).
medicine(migraine, painkiller).

% Doctors
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doctor(fever, dr_shah).
doctor(cough, dr_patel).
doctor(diabetes, dr_mehta).
doctor(migraine, dr_gupta).

% Rules

prescribe(Patient, Medicine) :-
   disease(Patient, Disease),
   medicine(Disease, Medicine).

consult(Patient, Doctor) :-
   disease(Patient, Disease),
   doctor(Disease, Doctor).

has_symptom(Patient, Symptom) :-
   disease(Patient, Disease),
   symptom(Disease, Symptom).
```

Queries (Enter one by one only)

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patient(X).

disease(rahul, X).

prescribe(amit, X).

consult(priya, X).

has_symptom(neha, X).

medicine(diabetes, X).

doctor(fever, X).

symptom(cough, X).

disease(X, migraine).

consult(rahul, X).
```

Features added

- 4 Patients
- 4 Diseases
- Symptoms
- Medicines
- Doctors
- 3 Rules (prescribe, consult, has_symptom)